COURSE PROJECT — FINAL REVIEW

AI-Driven Dual-Module System for Predicting Coated-Fertilizer Nutrient Release and Optimizing Crop Recommendations

Data Science Programming | CSI3001

Department of M.Tech Integrated Data Science

TEAM MEMBERS

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Problem Statement

Coated (controlled-release) urea is sold with a fixed release window, e.g. "90 days", measured in the lab at 25 °C. In the field, release speeds up in hot, wet soil and slows in cool, dry soil, so nitrogen arrives too early or too late for the crop.

Why a data-driven solution?

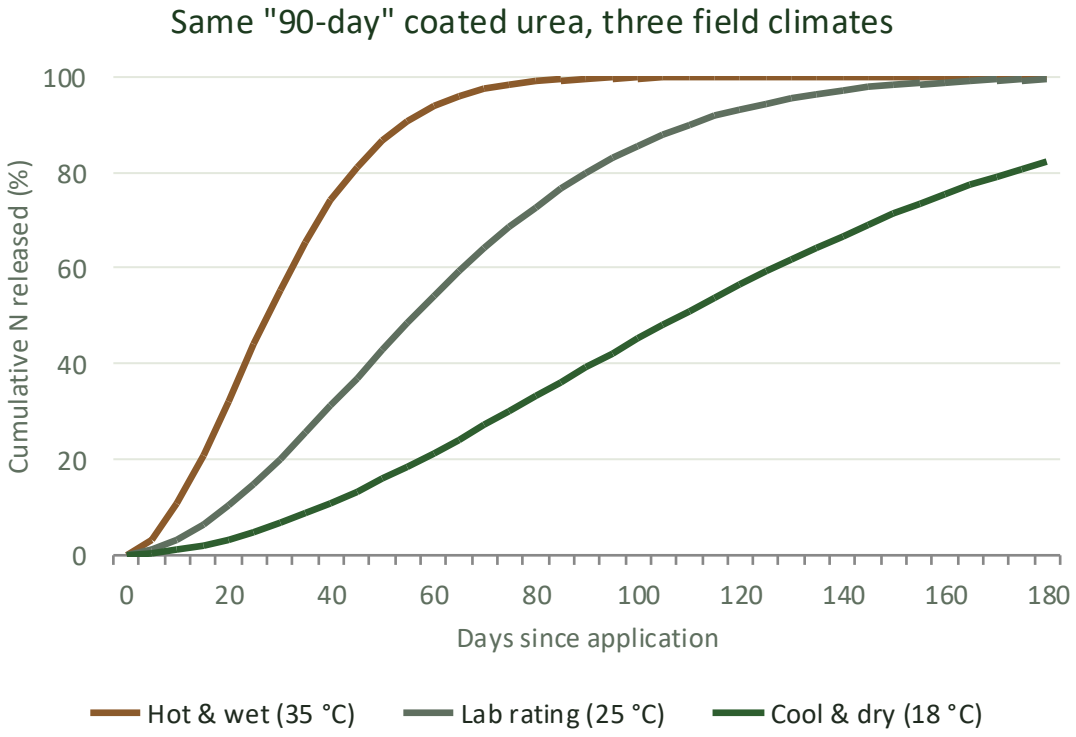
- Release depends non-linearly on temperature, moisture, rainfall and coating
- Soil-test charts give one static rate and ignore in-season weather
- Only 30–50% of applied nitrogen is taken up by crops

Who uses it

Farmers

Agronomists & extension officers

Fertilizer advisory & agri-tech apps



40–126 days

actual release time of a "90-day" product across 90% of the 11,000 field scenarios in our data

Objectives

	OBJECTIVE	MEASURED BY	RESULT
1	**Predict real-time nutrient release** from soil temperature, moisture, rainfall, pH and coating thickness	R^2 on the 20% test set	**R^2 0.98 – 0.99**
2	**Recommend the right fertilizer type** 7 products, including 3 coated-urea grades	Test accuracy (ceiling 88.4%)	**85.7% accuracy**
3	**Recommend amount and timing** kg/ha of product and days to the next application	R^2 on the 20% test set	**R^2 0.99 / 0.98**
4	**Synchronize release with crop demand** outperform the static 25 °C manufacturer rating	Coated-urea timing error (MAE)	**22.4 → 4.0 days**
5	**Deliver a usable tool** command-line predictor and notebook with input checks	Automated tests	**13 / 13 passing**

Proposed Solution

FIELD & FERTILIZER INPUTS

- Soil temperature
- Soil moisture
- Rainfall
- Soil pH
- Coating thickness

CROP INPUTS
Crop type · Growth stage · Initial N-P-K · Soil pH

MODULE 1

Nutrient Release Prediction

Polynomial LR · Decision Tree · Random Forest

Predicted release rate
+ synchrony index

MODULE 2

Fertilizer Recommendation Engine

XGBoost · step 1: type → step 2: amount & timing

OUTPUTS

- **Fertilizer type**
- **Amount (kg/ha)**
- **Timing (days)**

Practical outcome

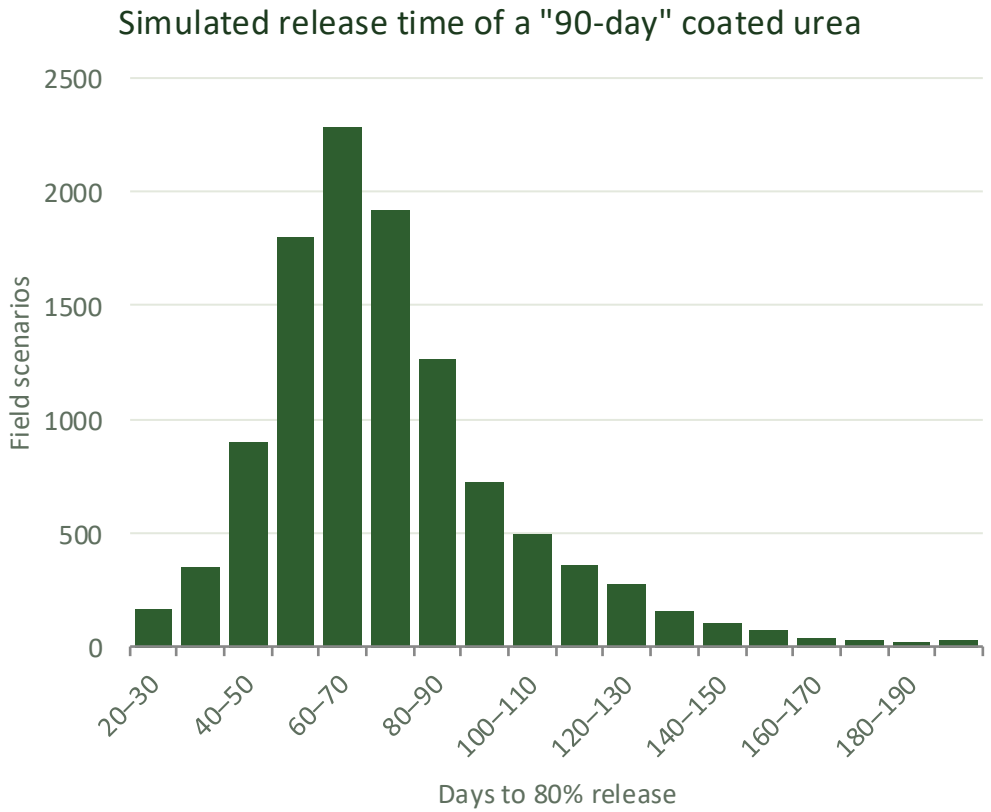
Right product, dose and date for each field, so nitrogen arrives when the crop needs it

Dataset

Dataset	Source	Size	Contents
Crop Recommendation	Real · Kaggle	2,200 × 8	N, P, K, pH, temperature, humidity, rainfall; 22 crops
Crop N-P-K requirements	Real · Harvestify	22 × 6	Target N, P, K per crop
Nutrient-release data	Generated · Module 1	11,000 × 11	6 field inputs → release rate, cumulative release
Recommendation data	Generated · Module 2	11,000 × 14	Crop, stage, N-P-K, pH, release → type, amount, timing

How release is simulated (no public field dataset exists)

- Each real field record → 5 season scenarios (soil temp, moisture, coating, days)
- Temperature: Q10 = 1.8, +10 °C ≈ halves release time (Brown et al., 1966)
- Coating: 90 days at 65 µm; dry soil slows release; S-shaped Weibull curve
- Noise: 6% batch variability, measurement noise, 3% label disagreement



Median 70 days; 90% of fields 40–126 days. The generated data reproduces the 50–120 day swing in our problem statement.

Data Preparation & Feature Engineering

Missing values

0 missing cells in all four datasets (checked with isna()). No imputation needed.

Duplicates

0 duplicate rows in the real Kaggle data and in both generated datasets.

Outliers

IQR rule flags high P, K and rainfall (e.g. apple and grape soils, K ≈ 200). Kept as genuine crop values; derived soil values clipped to 8–48 °C and 5–50%.

Encoding

One-hot: crop type (22), growth stage (4), fertilizer type (7). Fertilizer classes label-encoded for XGBoost.

Scaling

StandardScaler before linear and polynomial regression. Trees and XGBoost use raw values (scale-invariant).

Feature engineering

Soil temperature = air + 1–6 °C; soil moisture from humidity + rainfall; synchrony index = predicted release days ÷ crop days left.

Methodology

Model	Used in	Why it fits this problem
Linear / Polynomial Regression	Module 1	Interpretable baseline; degree-3 terms capture the exponential temperature effect
Decision Tree	Module 1; Module 2 baseline	Non-linear and explainable; gives feature importance
Random Forest	Module 1 benchmark; Module 2	Averages 300 trees → lower variance, robust to outliers
XGBoost	Module 2 (final)	Gradient-boosted trees; strongest on tabular data with feature interactions

TRAINING & VALIDATION PROCESS

1 80/20 train–test split (stratified by fertilizer type)

2 5-fold cross-validation on the 80% training set

3 Select the best model by mean CV score

4 Evaluate once on the held-out 20% test set

5 Compare with the static system and an ablation without Module 1

Metrics Regression: MSE · RMSE · MAE · R^2 Classification: accuracy · macro-F1 · confusion matrix

Model Implementation

Model	Key hyperparameters
Linear Regression	StandardScaler → ordinary least squares
Polynomial LR	PolynomialFeatures(degree = 3) → OLS
Decision Tree (M1)	GridSearchCV: max_depth {6, 10, 14, None}, min_samples_leaf {1, 5, 10} → best leaf = 5
Random Forest	300 trees; min_samples_leaf 1 (classifier) / 2 (regressor)
XGBoost classifier	400 trees, max_depth 6, learning_rate 0.08, subsample 0.9, colsample 0.9
XGBoost regressors	500 trees, max_depth 6, learning_rate 0.06 (amount & timing)
Decision Tree (M2)	max_depth 14, min_samples_leaf 3 (baseline)

Pipeline integration · src/pipeline.py

```
rate = module1.predict(field)          # % / day
X['predicted_release_rate'] = rate
X['synchrony_index'] = (80 / rate) / days_left

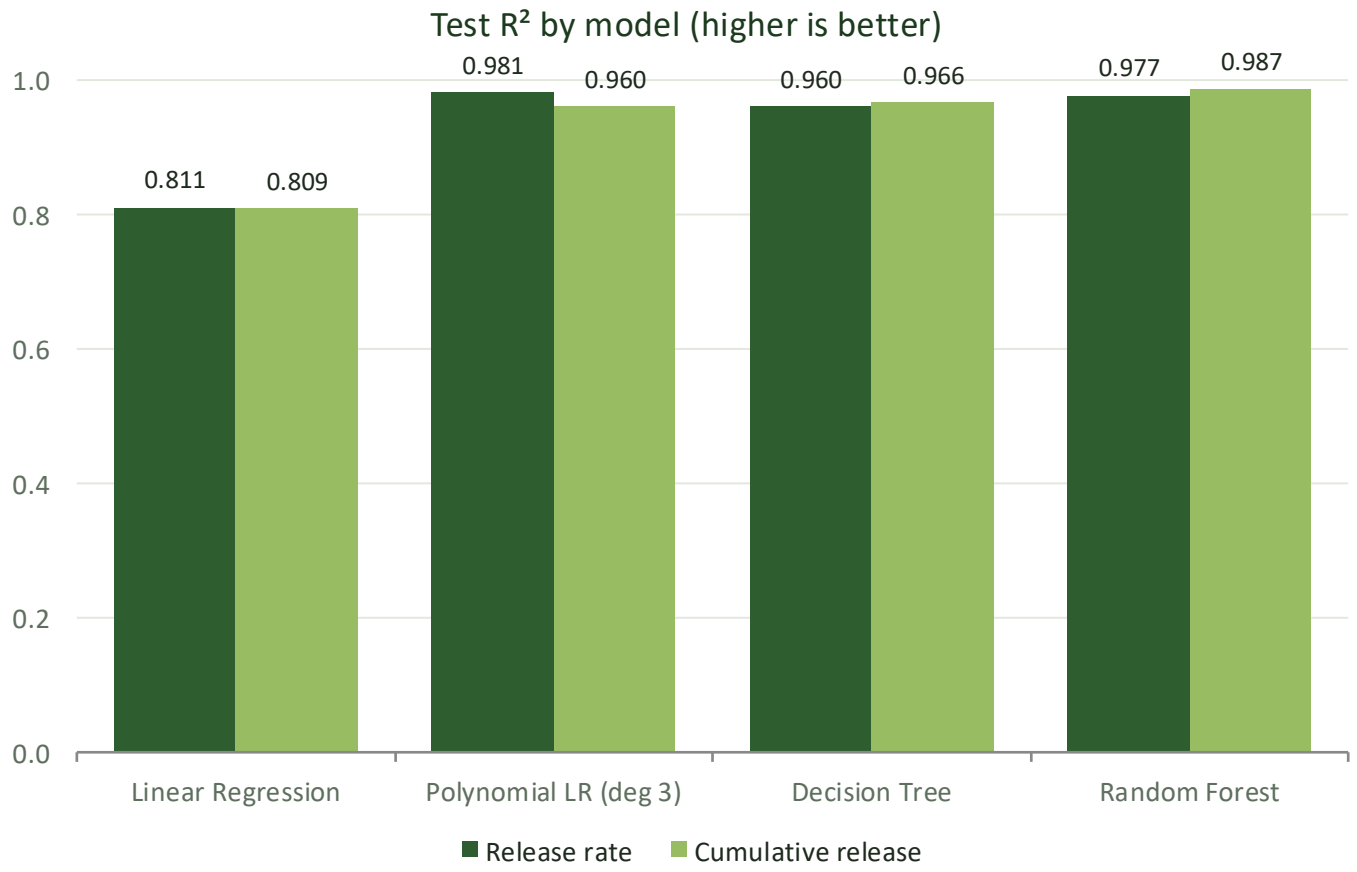
product = xgb_type.predict(X)          # step 1
X['fertilizer_type'] = product
amount = xgb_amount.predict(X)         # step 2
timing = xgb_timing.predict(X)
```

TOOLS & LIBRARIES

Python 3.9 · pandas 2.3 · NumPy 2.0 · scikit-learn 1.6 · XGBoost 2.1 · Matplotlib 3.9 · joblib · pytest

Project code src/ (8 modules) · main.py (end-to-end run, ~2 min) · predict.py (CLI app) · Smart_Fertilizer_Project.ipynb · tests/ (13 tests)

Results: Module 1, Nutrient Release



Release rate (% per day)

Best: Polynomial LR (deg 3)

R^2 0.981 · RMSE 0.124 · MAE 0.087

Cumulative release (%)

Best: Random Forest

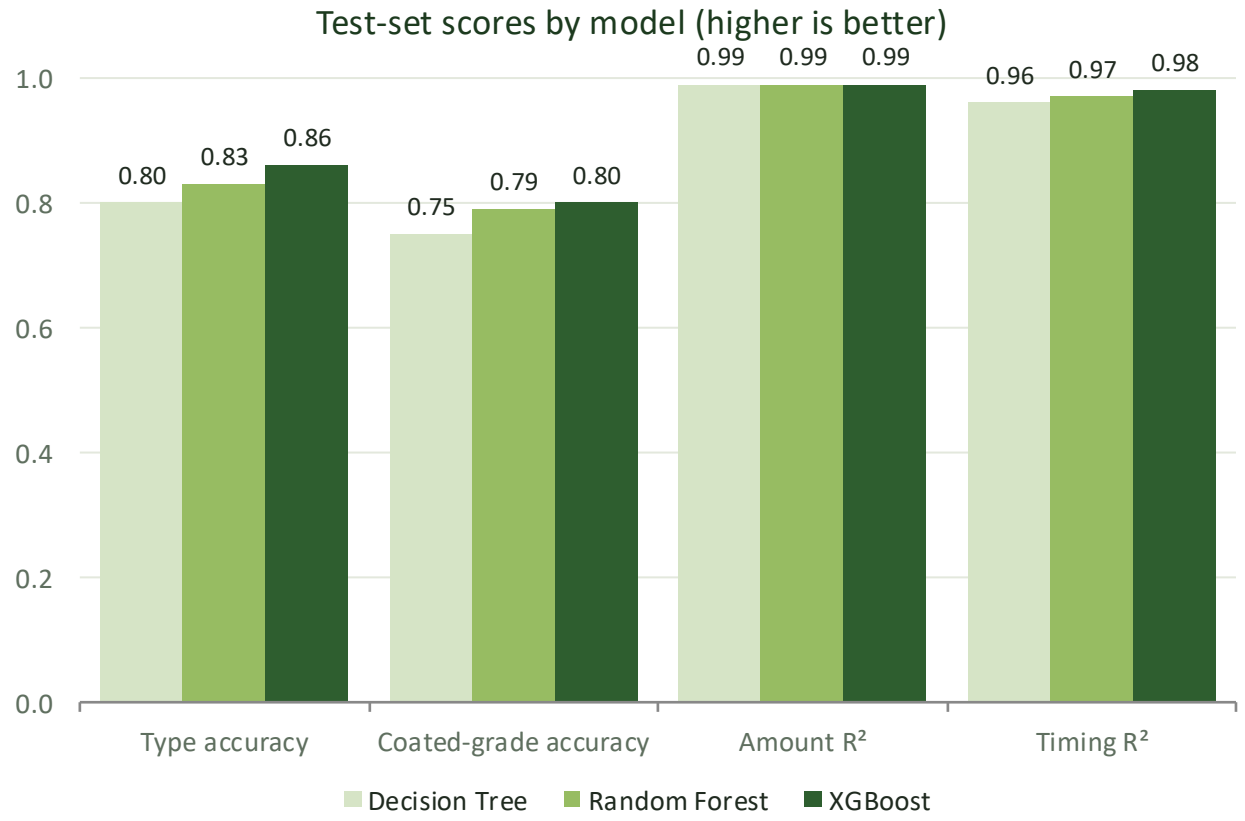
R^2 0.987 · RMSE 3.96 · MAE 2.66

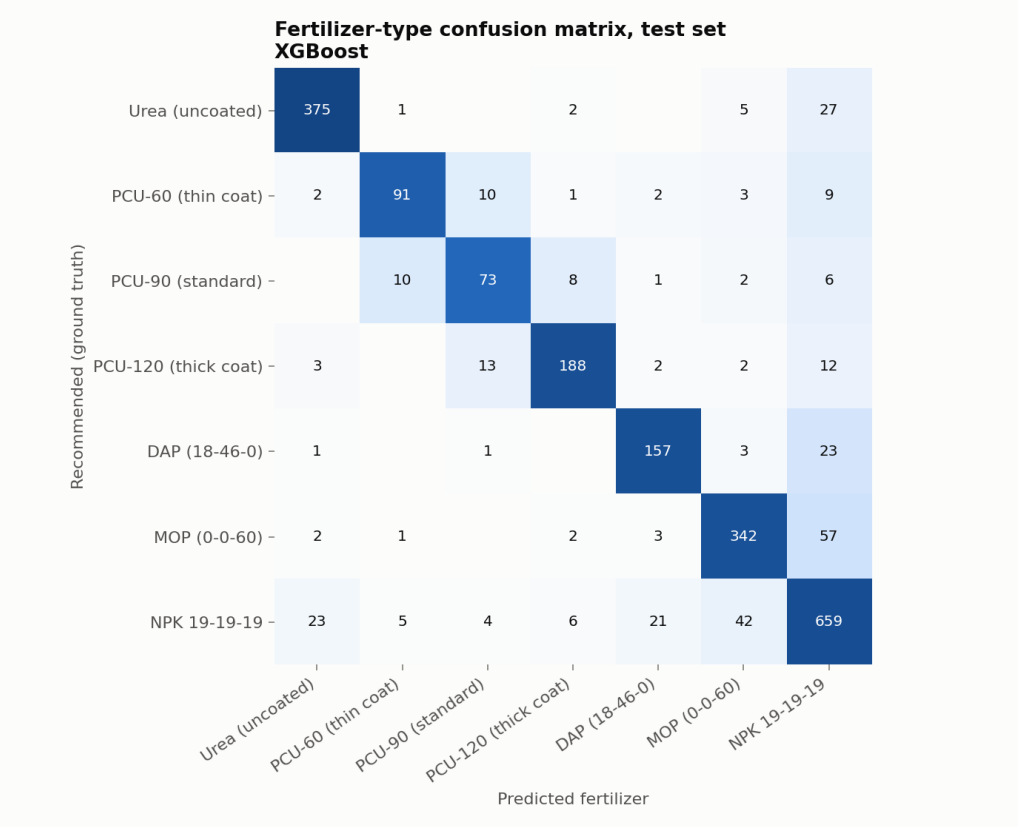
What the results show

- Plain linear regression underfits (R^2 0.811): temperature acts exponentially
- Coating thickness (57%) and soil temperature (40%) drive release-rate predictions

Evaluated on the 20% held-out test set (2,200 rows); best model chosen by 5-fold CV R^2.

Results: Module 2, Recommendation Engine





Best model: XGBoost. Highest 5-fold CV score on all three targets; 85.7% accuracy vs a 88.4% ceiling set by label noise.

Most errors are between neighbouring coat grades or with the general NPK product.

Results: Existing Static System vs Proposed

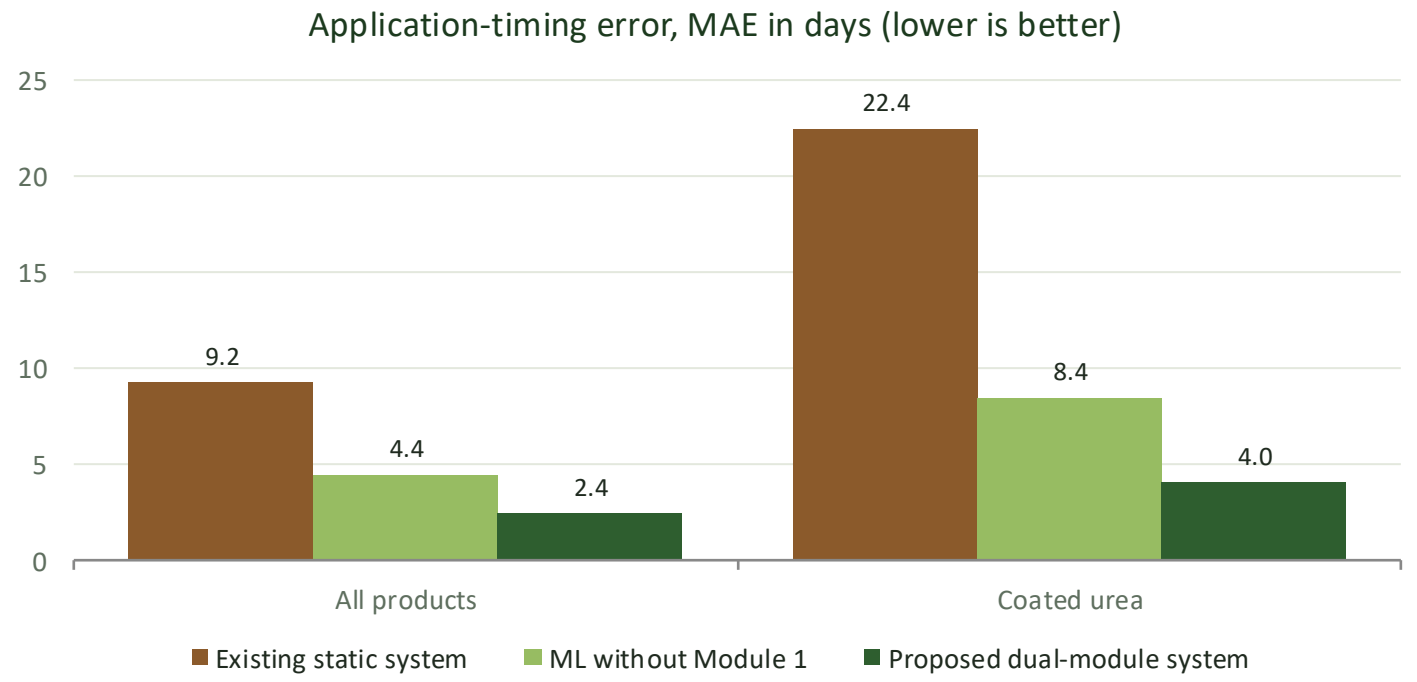
82%

lower timing error for coated urea than the manufacturer's static 25 °C rating (22.4 → 4.0 days)

Does Module 1 matter? (ablation)

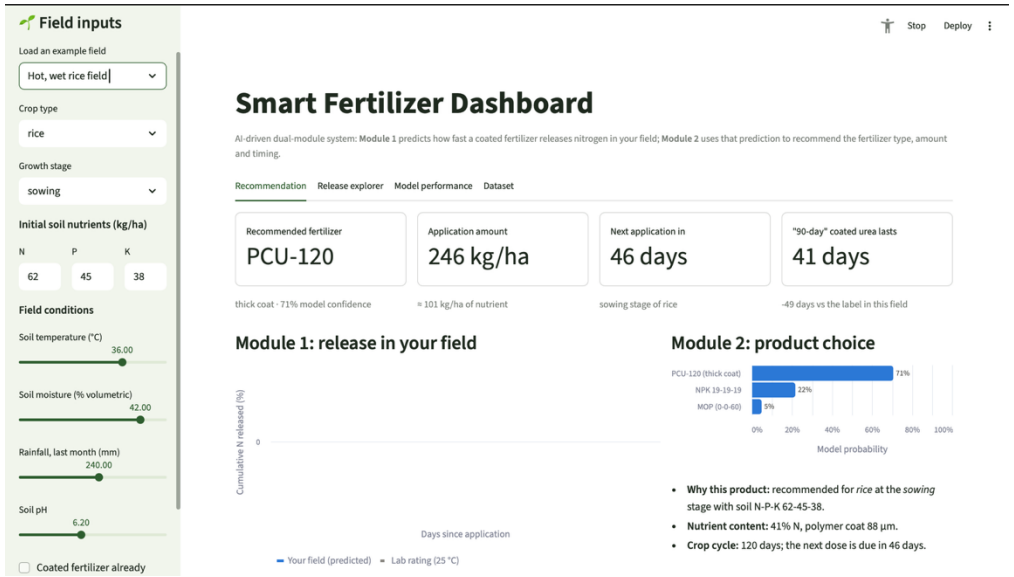
- Coated-grade accuracy: 71% → 80%
- Timing R^2: 0.90 → 0.98
- Type accuracy: 83.6% → 85.7%

Without → with Module 1's predicted release

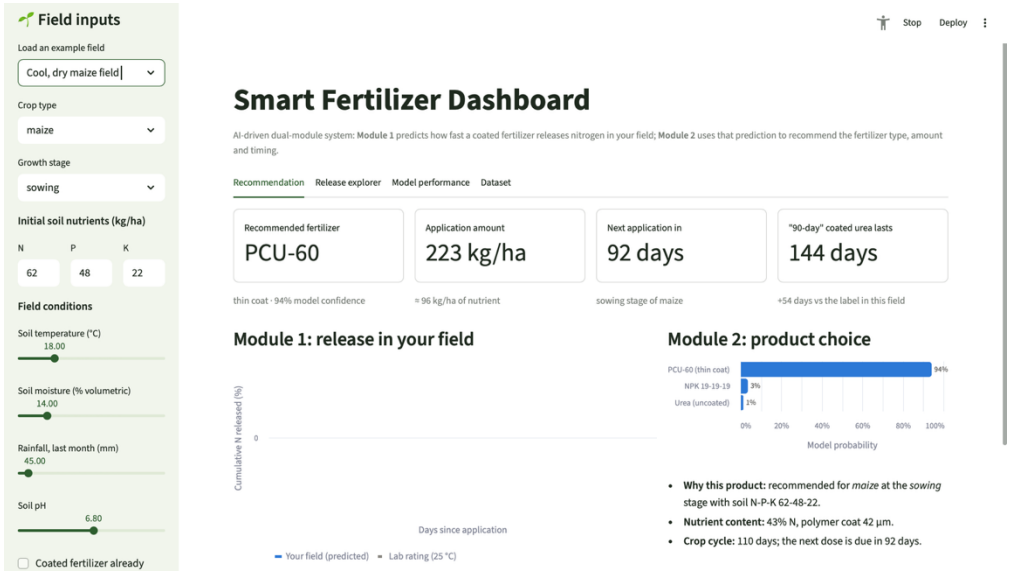


System	Timing MAE (all)	Timing MAE (coated)	Amount MAE
Existing static system	9.2 days	22.4 days	12.6 kg/ha
ML without Module 1	4.4 days	8.4 days	9.0 kg/ha
Proposed dual-module system	2.4 days	4.0 days	8.4 kg/ha

Application / Output



Hot, wet rice field (36 °C) → thick-coat PCU-120: a "90-day" product would run out in ~41 days



Cool, dry maize field (18 °C) → thin-coat PCU-60: only 10% released after 30 days

Before application

Advisor enters soil test and weather → product, dose and date

During the season

Re-run with current weather to reschedule the next top-dressing

Built-in safety

Inputs outside the training range are flagged as extrapolations

Challenges & Solutions

No public dataset of field release	Combined real Kaggle field data with literature-based release simulation
Dose depends on the product (amount R^2 only 0.77)	Chained models: predict type first, then amount and timing for it (R^2 0.99)
Unfair comparison with the static system	Evaluated every system on the same recommended product
Coat grades confused (~75% accuracy)	Engineered a synchrony index from Module 1's output (+3 points)
Risk of optimistic test scores	Selected models by 5-fold CV; test set used only once

Conclusion & Future Scope

KEY FINDINGS · ALL 5 OBJECTIVES ACHIEVED

- Module 1 predicts field release with R^2 0.98–0.99, replacing the fixed "90-day" label
- XGBoost picks the product with 85.7% accuracy (ceiling 88.4%) and sizes dose and timing with $R^2 \geq 0.98$
- Coated-urea timing error falls 82% vs the static rating (22.4 → 4.0 days)
- Module 1 is what makes the difference: removing it costs 9 points of coat-grade accuracy

Future scope

- Calibrate Module 1 with lysimeter or incubation release measurements
- Live weather-forecast API and IoT soil sensors
- More products: sulphur-coated urea, coated P & K, inhibitors
- Mobile app in regional languages for farmers
- Optimise for fertilizer cost and expected yield

Limitation: release behaviour and labels are simulated from published relationships; field validation is the next step.

References

DATASETS

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SOFTWARE & DOCUMENTATION

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